Using Clustering and PCA to Predict TV Show Attributes from Visual Features of Posters

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***Abstract*—This study aims to analyze genres in TV Posters through unsupervised learning, using Principal Component Analysis (PCA) and clustering models to determine patterns in visual features and metadata features. The results show that has the highest Silhouette Score (), Davies-Bouldin Index (), and the Calinski-Harabasz Index (). The genre is the superimposed to the clusters. External validation metrics will then be used to determine how genres are distributed and aligned to the clusters. The results indicate that the genre**

***Index Terms*—TV Posters, Genre Analysis, Unsupervised Learning, Principal Component Analysis, Clustering, Silhouette Score, Davies-Bouldin Index, Calinksi-Harabasz Index**

I. Introduction

Posters have been an effective marketing tool in the modern society. It is designed to evoke emotions, whether it will be for movies, advertisements, TV shows, or election campaigns. With the advancement of technology, it has now become possible to analyze these poster quantitatively by machine learning.

This study aims to assess the viability of using visual patterns and metadata of TV posters as predictors of genre, using Principal Component Analysis (PCA), and clustering methods such as K-means, Hierarchical, DBSCAN, Gaussian Mixture Models (GMM) and BIRCH. Lastly, it evaluates whether the clusters formed have a relationship with the genres by mapping and applying external validation metrics.

II. Review of Related Literature

As digital technology advances, the need to manage and utilize the information alongside it becomes vital [1]. Genre classification has been extensively studied for the past twenty years due to the advancement of technology. However, due to its nature of being a labeled data, most of the studies have been done by supervised learning methods.

A. Poster Analysis

Posters have been used throughout history since mankind is able to comprehend how to draw and color at places. Effective visual composition not only conveys narrative elements but also evokes specific emotions in viewers. For instance, the use of color theory within a composition can elicit feelings of joy, sadness, or tension [2].

There have been multiple studies that have been conducted. Majority are mostly done on movie posters and genre being the main feature to do a classification. For instance, the study of Andika and Wulandhari used Convolutional Neural Network to classify genres on films [3]. In another study, Chu and Guo applied Deep Neural Networks to the same problem [4].

B. Color-Based Feature Extraction

Color-Based Feature Extraction has been one of the techniques available when doing image classification. However, with the advancement of technology, more researchers have opted to use richer features such as edge detection, object extraction, etc.

A study has been done to classify genre based on the color features of movie posters using data transformation with distance ranking, kNN and Naives Bayes Classifier. (Ivasic-Kos, et al. 2014) [5]. Korai, et al., also did a similar study but with a different method called Deep Feed Forward Network [6].

C. Gaps in Literature and Study Contribution

Majority of the research has been done on movie posters. Most of the genre classification also has been done through supervised learning. This leaves a gap about understanding unsupervised learning in genre classification, but also being able to understand the properties of TV posters. This study provides a fresh perspective in the following:

- Due to genre being a labeled data, it will not be included in both the pre-processing PCA and clustering, ensuring an unbiased cluster group.
- The genre alignment will be done after clustering, giving insight as to whether TV poster form clusters themselves around a genre.

III. Methodology

A. Data Collection

The dataset gathered is scraped from IMDB, a publicly available website that contains extensive metadata from both films and tv shows. The total number of tv show posters downloaded is 99536. The metadata of tv shows has 59276 rows and 7 columns. The columns are "IMDB ID", "Title", "Rating", "Rating Content", "Date Published", and "Genres". The dataset will only utilize 50000 rows of metadata due to time constraints of processing images and the algorithms involved.

B. Data Pre-Processing

Incomplete, and null values were excluded from the metadata. Missing values were also excluded to ensure the quality of training and testing data. Red, Green, Blue and White mean values are extracted from the images using Computer Vision Techniques. Two variables were selected, "Rating and Date Published." The variables are chosen because they are numerical, which are the best features for PCA Analysis. The "Date Published" is truncated such that only the first four strings are selected as the variable is in a Year-Month-Date format. The focus is only "Year", hence only the first four strings is retrieved and converted into float.

C. Experimental Setup

This research utilized Python (Jupyter Notebook in Visual Studio Code) with libraries such as BeautifulSoup, Requests, Numpy, Pandas, CV2, Matplotlib, Seaborn, and Scikit-Learn. BeautifulSoup and Requests were used for scraping data. Pandas facilitated data manipulation. CV2 were used to extract image feature (RGBW mean values). Seaborn and Matplotlib were used for dataset visualization, and Scikit-Learn provides essential tools for model training and evaluation, including standard scaler, cross-validation, and error metrics.

D. Algorithms

The algorithms used in this research are:

1) K-Means Clustering: It clusters based on how near the data points are to a central chosen value [7]. The parameters used are $n_clusters = n_clusters$ (optimal value passed by the elbow score) and $random_state = 42$ (ensures consistency by having value).

2) Hierarchical Clustering: It clusters based on the proximity distance (degree of relationship) between objects [7]. The parameter linkage used are (to use dendrogram) with the dataset features to cluster and $method = ward$ to minimize the variance in clusters. The parameter fcluster has parameters Z (output of linkage), n-clusters (the number of clusters to form, in this case from two to seven), and $criterion =' maxclust'$ (forcing the exactly the number of clusters).

3) DBSCAN: It clusters based on the density of data points on a certain distance [7]. The parameters used are $ep = [0.3, 0.5, 0.7, 1.0]$ The basis being what was saw in scatter graph of PC1 and PC2 being highly dense. The $min_samples = 5$, the minimum number of point to form a cluster.

4) GMM: It clusters based on the assumption that the data points is on a Gaussian Distribution and tries to guess the probability of the location of the point part of a cluster [7]. The parameters are $n_components = n_clusters$, $random_state = 42$.

5) BIRCH: It clusters in a Feature Tree. It works better in large datasets [7]. The parameters used are $n_clusters = n_clusters$ (optimal value passed).

E. Training Procedure

For the training model, the researcher implemented feature engineering to transform one categorical label, "Genre", into binary values using One-Hot Encoding to be used for mapping post-clustering. The numerical features are then Standardized through StandardScaler. Principal Component Analysis (PCA) was applied to reduce dimensions, retain key components, and reject redundant data.

F. Evaluation Metrics

1) For the Performance of Clustering Algorithms: The study uses three evaluation metrics to evaluate each algorithm.

- Silhouette Score - measures how well separated clusters are. Given by the formula

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{1}$$

- Davies-Bouldin Index - measures cluster compactness and separation. Given by the formula

$$\text{DBI} = \frac{1}{N} \sum_{i=1}^{N} \max_{j \neq i} \left(\frac{s_i + s_j}{d_{ij}} \right) \tag{2}$$

- Calinski-Harabasz Index - measures the ratio of between-cluster dispersion and within-cluster dispersion. Given by the formula

$$CHI = \frac{between - clusterdispersion}{within - clusterdispersion} \tag{3}$$

2) For the Validation of the Genre Distribution: The study uses three external validation metrics to determine how well aligned the clusters are with the actual categories of genre.

- Purity Score - measures the percentage of the cluster us dominated by one genre. Given by the formula

$$Purity = \frac{1}{N} \sum_{i=1}^{K} \max_{j} |C_i \cap G_j| \tag{4}$$

- Normalized Mutual Information - Measures how much genre information is shared with clusters (mutual dependence). Given by the formula

$$NMI = \frac{2I(C, G)}{H(C) + H(G)} \tag{5}$$

- Adjusted Rand Index - Measures the similarity between clusters and actual genres, adjusting for randomness. Given by the formula

$$ARI = \frac{\sum_{ij}\binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{N}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]/\binom{N}{2}} \quad (6)$$

IV. Results and Discussion

A. Principal Component Analysis

In this study, PCA was used to analyze the relationship between RGBW mean values, and the metadata variables year and rating. It was also used to determine the best features to use in the model.

Principal Component	Explained Variance Ratio
PC1	0.5443
PC2	0.1747
PC3	0.1580

TABLE I
Explained Variance Ratio

In Table 1, three components, PC1, PC2, and PC3 are needed to reach the minimum threshold of 80 percent according to the Kaiser Criterion and Scree Plot Interpretation.

Feature	PC1	PC2	PC3
r_mean	0.493097	-0.011518	0.006017
g_mean	0.529801	-0.021074	0.027183
b_mean	0.497853	0.013930	0.030935
w_mean	0.475757	0.016880	0.024097
rating	-0.032752	-0.704153	0.708996
year	-0.029866	0.709304	0.703571

TABLE II
Feature Loadings for Principal Components

In Table 2, PC1 exhibited significant scores on the RGBW mean values. PC2 showed negative relationship between rating and year. PC3 showed that rating and year are positive relationship.

B. Clustering Analysis

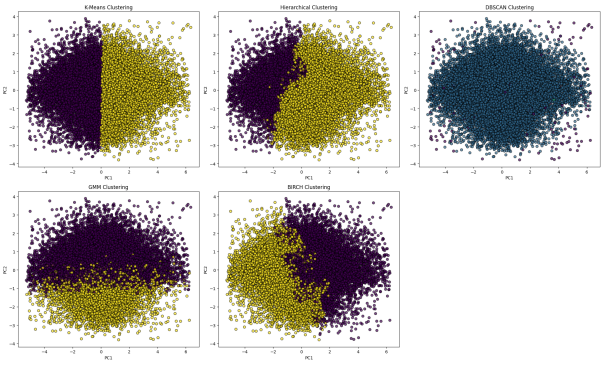


Fig. 1. Comparison of Different Clustering Algorithms.

Figure 1 shows the results of the clustering algorithms after PCA were performed in the feature dataset. PC1 represents the x-axis while PC2 represents the y-axis.Besides DBSCAN, every algorithm has 2 clusters. K-means (first one from the left) shows a clear hard line in the middle when PC1 is zero. Hierarchical has somewhat similar in structure to K-means but breaks down and skews to the negative PC1 when PC2 is negative. The opposite happens with BIRCH where when PC2 gets negative, PC1 gets slightly positive. GMM Clustering however is similar to K-means except when PC2 is zero. DBSCAN just has one cluster which is understandable since its based on density.

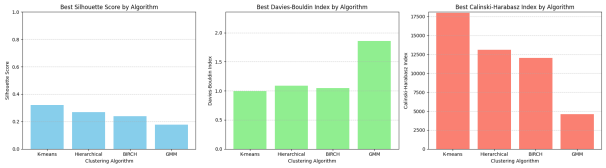


Fig. 2. Evaluation Metric of Different Clustering Algorithms.

Figure 2 shows the results of evaluation metrics for the clustering algorithms. K-means outperforms the other three algorithms, with a Silhouette Score of 0.322, DBI of 0.996, and CHI of 18008.055.

C. Validation Metrics

After clustering, the genre dataset will then be mapped into the clusters and do a validation metric to determine whether the clusters are aligned with the genre.

Metric	Score
Normalized Mutual Information (NMI)	0.019
Adjusted Rand Index (ARI)	0.008
Purity Score	0.337

TABLE III
Validation Metric Against Genres

As shown in Table 3, the NMI score being low suggests limited alignment between clusters and genre. Same result with the ARI. Purity Score being slightly lower than 0.5 mean its a bit more of a mix of genres rather than being a genre specific cluster.

V. Conclusion

This research has explored the effectiveness of doing a genre mapping after an unsupervised learning. Through the use of Principal Component Analysis (PCA) and clustering algorithms, the study established significant categories, allowing for effective targeted marketing when making tv show posters. The combination of K-Means, Hierarchical, DBSCAN, Gaussian Mixture Models (GMM) and BIRCH on the relationship between the visual features of the poster and the year and ratings, with K-Means being the best clustering algorithm according to the Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.

The research confirmed that, with intense data preprocessing, feature engineering, and model assessment, genre

mapping is not an effective tool for companies to use in determine genre when making posters. However the results validated that machine learning-based trend analysis can greatly enhance the marketing capabilities to determine whether the visual features alongside year and ratings can predict the genre of the tv show poster.

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